Week 2

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Question 1

In [9]:
```
import statistics

class_marks=[65, 72, 80, 55,90, 68, 75, 82, 60, 78]

mean_mark = statistics.mean(class_marks)
median_mark= statistics.median(class_marks)
std_mark= statistics.pstdev(class_marks)

print("Mean:", mean_mark)
print("Median:", median_mark)
print("Population Standard Deviation:", std_mark)
```

```
Mean: 72.5
Median: 73.5
Population Standard Deviation: 10.141498903022176
```

Question 2

In [21]:
```
tank_A = int(100)
tank_B = int(200)

print ("Type of tank A:", type(tank_A))
print ("Type of tank B:", type(tank_B))
```

```
Type of tank A: <class 'int'>
Type of tank B: <class 'int'>
```

In [22]:
```
total_volume=tank_A + tank_B
print ("Total volume:", total_volume, "L")
```

```
Total volume: 300 L
```

In [23]:
```
tank_A = str (100)
tank_B = str (200)

print ("Case 2")
print ("Type of tank A:", type(tank_A))
print ("Type of tank B:", type(tank_B))
```

```
Case 2
Type of tank A: <class 'str'>
Type of tank B: <class 'str'>
```

In [24]:
```
total_volume = tank_A + tank_B
print ("Total volume:", total_volume, "L")
```

```
Total volume: 100200 L
```

Question 3

In [10]:
```
def sum(a,b):
    return a+b
print(sum(6,7))
```

```
13
```

In [9]:
```
def sub(a,b):
    return a-b
print(sub(10,5))
```

```
5
```

In [8]:
```
def mult(a,b):
    return a*b
print (mult (2,3))
```

```
6
```

In [7]:
```
def divi(a,b):
    return a/b
print(divi (14,7))
```

```
2.0
```

In [13]:
```
def funct1(a,b):
    c = 10
    return (a+b)*c  #sam as return mult(sum(a,b),c)
print(funct1(4,5))
```

```
90
```

Question 4

In [2]:
```
subjects = ["Software", "Chemistry", "English", "Arabic"]

def subject_availability(subject_name):
    if(is_available(subject_name)):
        return "the subject is available"
    else:
        return "the subject is not available"
def is_available(subject_name):
    return subject_name in subjects #will returns either true or false
```

In [3]:
```
input_subject = input("Enter the name of the subject: ")

print (subject_availability (input_subject))
```

```
the subject is not available
```